Penchala Siva Prasanth Kasa

pkasa@wisc.edu | +1 (608) 236-3062 | LinkedIn | GitHub | Portfolio

Available for full-time employment from June 2027 (M.S. completes May 2027).

Summary

Software engineer with 2+ years of production **C/C++** and **Python** experience across embedded systems, ML infrastructure and backend services, now an M.S. in CS at UW–Madison. Shipped a metering device's software stack end to end at Enphase, used **Python** to benchmark and tune a heterogeneous inference runtime at Qualcomm, and automated production **ML** scoring at Visa. Comfortable across systems, distributed backends, and full-stack web. Dual degree from IIT Kharagpur; **JEE Mains Paper-II All-India Rank 23**.

Experience

AI Software Intern @ Qualcomm, Inc., *San Diego, USA* May 2026 - Aug. 2026

- Wrote **Python** benchmarking and profiling harnesses for a heterogeneous inference runtime co-executing on GPU and HTP.
- Analyzed cross-device data movement and sync in Python to cut latency; findings drove cost-aware routing of operators.
- Contributed C++ to Qualcomm's ExecuTorch backend, including graph-partitioning logic for operator placement.

Software Engineer @ Enphase Energy, *Bangalore, India | 2 years, full-time* Jul. 2023 - Jul. 2025

- Owned the **software development lifecycle** for a new metering device: design, **C/C++** build, code review, test, release.
- Wrote the stack ground up, bootloader through application, on Mbed OS/Zephyr; shipped to devices at fleet scale.
- Cut RAM 98% to 82% and storage by 20% by redesigning data structures, memory layouts and data paths fleet-wide.
- Hardened multithreaded C/C++ services and Protobuf telemetry encoding for real-time transport over CAN and Zigbee.
- Triaged live production failures from sparse field evidence with gdb and JTAG; drove root-cause and design reviews.
- Scripted/automated build, flash and regression checks in Bash/Python, gating safe deployment of fleet releases.

Software Intern @ Visa, Inc., *Bangalore, India* May 2022 - Jul. 2022

- Shipped **sub-10ms** ML scoring into a production pipeline handling **millions of transactions daily**, with safe fallbacks.
- Automated model onboarding with Go/Java microservices over Kafka and Spark, saving 2 weeks of manual work per model.
- Centralized scattered YAML model and alert configs into a MySQL schema, improving monitoring and data analysis.

Selected Projects

NoFi - Offline Peer-to-Peer Web App | TypeScript, React | MADHACKS, *UW-Madison* Nov. 2025

- Shipped a **TypeScript/React** browser app sending text and voice over audio frequencies (FSK) with no network.
- Wrote the real-time encoder/decoder against the Web Audio API behind a UI state machine that kept the app responsive.

Geospatial Real-Time Prediction & Fault-Tolerant Serving | *UW-Madison* Mar. 2024 - Oct. 2024

- Built a real-time serving pipeline (CNN + TabNet) over multi-sensor signals, serving predictions under latency targets.
- Implemented failover so critical paths kept serving through partial outages; live **WebSocket** dashboard and alerting.

VecScale - Distributed Vector Search (C++) | High Performance Computing, *UW-Madison* Jan. 2026 - May 2026

- Built a distributed top-k search system in C++ with MPI across nodes, OpenMP within nodes and CUDA on GPU.
- Measured strong/weak scaling, latency and recall@k; CMake and Slurm scripts for reproducible benchmark runs.

Job Board Crawler & Change Detector (Python) | *Personal* 2025 - 2026

- Built a crawler that scans **thousands of postings across 62 job boards** nightly and emails only the new matches.
- Isolated per-source failures behind retries, jittered rate limits and structured logs, so one board never halts a run.

Technical Skills

- **Languages:** C++, C, Python, Java, JavaScript/TypeScript, Go, SQL, Bash, HTML/CSS
- **Systems & Embedded:** Linux/UNIX, RTOS (Mbed OS/Zephyr), multithreading & concurrency, memory optimization
- **Web & Full-Stack:** React, REST APIs, WebSockets, responsive UI, Web Audio API
- **Distributed & Cloud:** AWS, Azure, Docker, microservices, Kafka, Spark, MPI/OpenMP/CUDA
- **Tooling:** Git, CMake/Make, gdb, JTAG, code review, CI/CD, Agile/Jira, MySQL/SQLite

Education

University of Wisconsin, Madison, USA Sep. 2025 - May 2027

Master of Science(MS) in Computer Science *School of Computer, Data & Information Sciences*

- **Coursework:** Operating Systems, High Performance Computing, Databases, Big Data Systems, AI Agents.
- **Graduate Teaching Assistant** for CS 640: Computer Networks.

Indian Institute of Technology, Kharagpur, India July. 2018 - May 2023

Dual Degree - B.Tech(Hons.) + M.Tech *Electronics & Electrical Communication Engineering*

- **Coursework:** Data Structures & Algorithms, Computer Networking, Embedded Systems Design.

Achievements

- **Nationally Ranked 23** among 125,000 candidates for IIT JEE Mains Paper-II (2018). (Top 0.018%)
- Nationally Ranked 1318 among 230,000 candidates for IIT JEE Advanced (2018). (Top 0.5%)
- Nationally Ranked 1250 in the Graduate Aptitude Test in Engineering (GATE - 2022). (Top 1%)